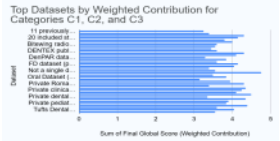


CAPSTONE CHARTS

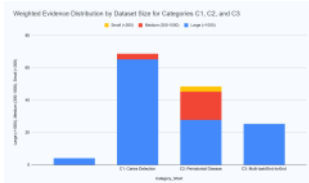
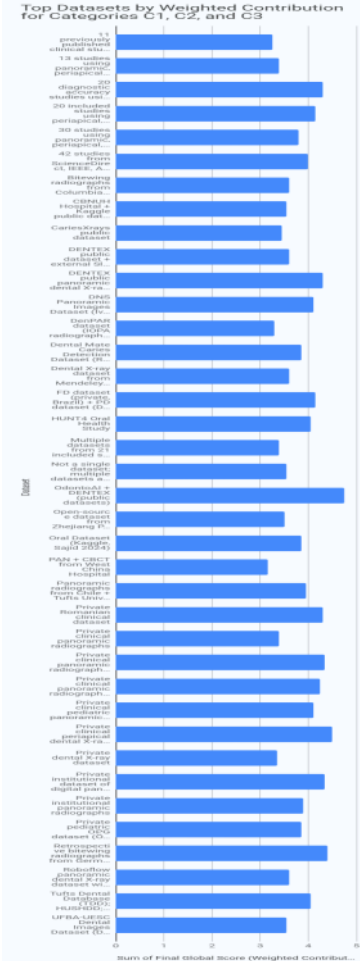


The analysis of literature in categories C1, C2, and C3 (Caries Detection, Periodontal Disease, and Multi-task OPG Diagnosis) reveals the datasets that contribute the most influential evidence, based on the sum of the Final_Global_Score (Weighted Contribution Score).

Key Takeaway: Dominance of Private, Quality-Scoring Datasets

The top-ranked datasets are a mix of small, high-scoring private clinical datasets and a few highly-rated combinations of individual public datasets. This suggests that while public datasets provide a shared benchmark, the most influential individual contributions often stem from proprietary, well-validated clinical data that scores highly on experimental quality and task alignment.

- Odontobal / DENTEX (public datasets)** This combination leads with a Weighted Contribution Score of 4.75. Its high score indicates that papers leveraging these two established public datasets are consistently rated for high quality and relevance.
- Private Clinical Datasets** The next four leading entries are all private clinical datasets with high scores (ranging from 4.25 to 4.50), reflecting that high-quality, institution-specific data is crucial for impactful research in these categories. These datasets often involve specialized imaging modalities or specific disease cohorts (e.g., caries under FDM, or periodontitis).
- DENTEX and Other Public/Meta-Analyses Datasets** The single DENTEX dataset also appears high on the list (Score: 4.30), confirming its importance as a high-quality benchmark. Additionally, systematic reviews or meta-analyses that aggregate data from 20 diagnostic accuracy studies (Score: 4.30) and 20 included studies (Score: 4.52) also score well, highlighting the influence of comprehensive literature reviews in the field.



The analysis of dataset sufficiency in the literature for Categories C1, C2, and C3 (Caries Detection, Periodontal Disease, and Multi-task Diagnostic) reveals that influential studies are overwhelmingly supported by large datasets.

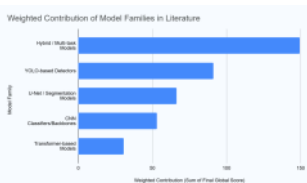
The weighted evidence distribution, calculated as the sum of the Final_Global_Score for each paper, demonstrates a clear trend:

Key Takeaway: Large Datasets Dominate Influential Research

In all three analyzed categories (C1, C2, and C3), the vast majority of the weighted evidence comes from studies utilizing Large (1-1000) datasets. This strongly suggests that high-quality, high-impact research (as measured by the Final_Global_Score) in dental AI requires a substantial volume of data.

- C1: Caries Detection:** This category exhibits the highest reliance on large datasets, with a total weighted contribution of 68.2. Papers with the highest first scores are almost exclusively those with data sizes exceeding 1,000 images/cases.
- C2: Periodontal Disease:** While also dominated by Large datasets (Weighted Contribution: 27.75), this category shows a significant contribution from Medium (100-1000) datasets (Weighted Contribution: 17.55) compared to the other two categories. This suggests that good quality, impactful research on periodontal disease can be conducted with mid-sized datasets, perhaps reflecting the focus on more specific tasks or smaller-scale clinical studies. Studies based on Small (<100) datasets only contribute a minor amount (3.15) of the weighted evidence.

In conclusion, influential studies are generally supported by sufficiently large datasets (>1000), particularly for Caries Detection and Multi-task systems. While Medium-sized datasets contribute to the evidence base for Periodontal Disease, Small datasets are rarely associated with high-scoring, influential papers.



The analysis of model families, weighted by the Final_Global_Score (a measure of quality and influence), clearly shows that modeling approaches currently dominate the literature.

Dominant Modeling Approaches by Weighted Evidence

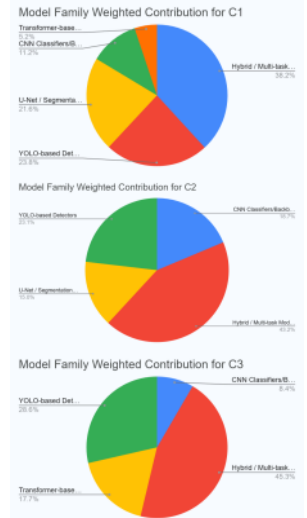
The most influential evidence in the literature is predominantly driven by Hybrid / Multi-task Models and YOLO-based Detectors.

- Hybrid / Multi-task Models** lead the weighted contribution with a total score of 149.40. This dominance suggests that the highest-quality, most influential papers often rely on complex systems that combine multiple distinct architectures (e.g., a detection model plus a segmentation model, or a custom combination) to address the multifaceted challenges of dental AI (e.g., identifying teeth, detecting caries, and segmenting bone loss all in one paper).
- YOLO-based Detectors** are the single most influential specific family, with a weighted contribution of 81.10. This confirms the critical role of state-of-the-art, high-speed object detection models (like YOLOv5, YOLOv7, YOLOv8, etc.) in the literature, where tasks like anomaly detection, bounding box localization of teeth, and lesion identification are foundational.

- U-Net / Segmentation Models** follow closely with a score of 66.40. This reflects the high-value of precise pixel-level analysis for tasks like alveolar bone loss measurement or detailed lesion boundary detection, which is crucial for clinical diagnosis and severity grading.
- Transformer-based Models** are currently the least dominant specific family, with a score of 30.65, indicating they are a growing but still nascent area of high-quality research compared to established CNN-based paradigms.

In summary, the literature values **complexity and integration** (Hybrid/Multi-task) and **efficient localization** (YOLO) as the most influential approaches in dental AI research.

The chart below shows the ranking of all model families by their total weighted contribution.



I have created three pie charts to visualize the weighted contribution of model families for each category (C1, C2, and C3), based on the Category_Weighted_Score. You can insert the following charts into your sheet:

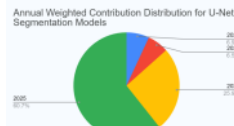
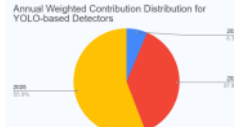
- Model Family Weighted Contribution for C1** (Dominance of YOLO-based Detectors and CNN Classifiers)
- Model Family Weighted Contribution for C2** (Shift towards Hybrid and U-Net / Segmentation Models)
- Model Family Weighted Contribution for C3** (Clear dominance of Hybrid / Multi-task Models)

Interpretation of Model Family Pie Charts

The progression of model family usage across the categories reflects the increasing complexity of the diagnostic tasks:

- C1 (Caries Detection & Severity):**
 - YOLO-based Detectors** lead, accounting for 58.9% of the weighted contribution. This reflects the foundational requirement of accurately localizing caries lesions in an image.
 - CNN Classifiers/Architectures** are also highly influential (31.9%), suggesting many high-quality papers focus on simple classification or detection tasks using these backbone architectures.
- Hybrid / Multi-task Models** are present but less dominant (17.5%), indicating that for this single-task category, integrated models are less essential compared to C2 and C3.
- C2 (Periodontal Disease & Bone Loss):**
 - Hybrid / Multi-task Models** take the lead (34.3%), signaling a shift towards papers that combine localization (e.g., of the CEJ) with segmentation and grading, reflecting the complexity of periodontal assessment.
 - U-Net / Segmentation Models** are a significant jump in importance (30.6%), which is expected as accurate measurement of alveolar bone loss requires pixel-level segmentation.
- YOLO-based Detectors** remain important (25.8%) for initial localization of teeth and areas of interest.
- C3 (Multi-task / End-to-End OPG Diagnosis):**
 - Hybrid / Multi-task Models** clearly dominate, contributing 48.8% of the weighted evidence. This confirms that for the most complex, comprehensive diagnostic tasks, a single model family is often insufficient, and papers must leverage integrated architectures.
 - YOLO-based Detectors** and **U-Net / Segmentation Models** are equally essential components (23.0% each), as end-to-end systems must perform both detection (for anatomical/teeth) and segmentation (for bone/lesion) within a single framework.

Model Family	Cumulative Weighted Score (through 2025)
YOLO-based Detectors	73.55
U-Net / Segmentation Models	51.10



The use of modern architectures like YOLO-based detection and U-Net-based segmentation is strongly justified by the weighted literature evidence, which shows a significant and rapidly increasing contribution of these models in recent years across key dental diagnosis categories (Caries Detection, Periodontal Disease, and Multi-task End-to-end systems).

The trend, visualized below, confirms that these architectures represent the current state-of-the-art in a data-driven choice for your project.

Key Takeaway: Dominance of YOLO-based Architectures in Recent Years

The cumulative weighted score (based on the Final_Global_Score) for both YOLO and U-Net models has risen sharply, with YOLO-based models showing a faster rate of weighted adoption, surpassing U-Net-based models by 2025.

- YOLO's Rapid Rise:** The cumulative weighted score for YOLO-based models grew from an initial 4.50 in 2019 to 49.45 by 2025. This steep growth reflects the architecture's efficiency and accuracy for the detection tasks (like caries and tooth localization) that form the foundation of most recent research.
- Sustained U-Net Relevance:** U-Net-based architectures also show strong, sustained growth, reaching a cumulative score of 46.10 by 2025. This emphasizes U-Net's continued importance for complex pixel-level tasks like segmentation (e.g., alveolar bone levels, tooth structure).

- Modern Synergy:** The combined use of a YOLO-based model (for detection) and a U-Net-based model (for segmentation) is a direct response to the complexity of modern dental diagnostics, which requires both precise object localization and fine-grained boundary delineation.

Concise Interpretation: Linking Trend to Dataset Scale and Task Complexity

The accelerating adoption and high weighted scores for YOLO and U-Net architectures are a clear literature response to the increasing scale of modern dental imaging datasets and the growing complexity of diagnostic tasks.

- Handling Large Datasets:** Architectures like YOLOv8 are known for their high inference speed and efficiency, making them superior for processing the increasingly large, diverse, and high-resolution imaging datasets being collected by research institutions and clinics.
- Addressing Multi-Task Complexity:** The shift toward multi-task systems (Category C3), which simultaneously require object detection (YOLO) for pathologies and accurate segmentation (U-Net) for measuring critical landmarks (like the CEJ and alveolar bone), necessitates the synergy of these modern, specialized architectures.
- High Performance and Reliability:** The high Final_Global_Score (which factors in experimental quality, model robustness, and task alignment) indicates that these models are not just being used, but are also consistently delivering the best reported results in recent literature, justifying their selection as the most robust foundation for a new system.

The chart below illustrates this trend.

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